

Configure a LAN (Local Area Network)

AIM: To setup and configure a LAN (Local Area Network) using a network switch and Ethernet cables in a computer laboratory.

What is a LAN?

A Local Area Network (LAN) is a computer network that connects devices within a limited geographical area such as a home, school, office, or computer laboratory. It allows connected devices to communicate with each other and share resources such as files, printers, applications, and internet access. A LAN provides fast and reliable communication between computers and other network devices.

A network switch is commonly used as the central connecting device in a wired LAN. It receives data from one device and forwards it to the appropriate destination device. This helps computers on the same network communicate efficiently.

Equipment Required

- 4 Computers or PCs
- 1 Ethernet network switch (8-port, 16-port, or 24-port)
- 4 Ethernet cables (Cat5e/Cat6)
- Network interface cards (NICs) available on the computers

Network Topology

A star topology is used for this LAN setup. All computers are connected individually to the central network switch using Ethernet cables.

Procedure to Set Up a LAN

Step 1 – Plan the Network: Plan the network topology and decide the location of the computers and network switch. Use a star topology in which the switch acts as the central connecting device.

Step 2 – Prepare the Equipment: Arrange four computers, one Ethernet switch, and four Ethernet cables. Ensure that the network ports on all computers and the switch are working properly.

Step 3 – Connect the Computers to the Switch: Connect one end of an Ethernet cable to the Ethernet/LAN port of each computer and connect the other end to a port on the network switch. Repeat the process for all four computers.

Step 4 – Configure IP Addresses: Assign a unique IPv4 address to each computer. All computers should belong to the same network/subnet.

Step 5 – Open Network Settings: On Windows, open Control Panel → Network and Internet → Network and Sharing Center → Change adapter settings. Right-click Ethernet and select Properties.

Step 6 – Configure IPv4: Select Internet Protocol Version 4 (TCP/IPv4), click Properties, select 'Use the following IP address', and enter the appropriate IP address, subnet mask, and other required settings.

Step 7 – Verify the Connection: Open Command Prompt and use the ping command to test communication between the computers. A successful ping indicates that the computers can communicate over the LAN.

Sample IP Address Configuration

Computer	IPv4 Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0
PC4	192.168.1.4	255.255.255.0

Testing the LAN Connection

1. Open Command Prompt on any computer.
2. Use the following command to test connectivity with another computer:

ping 192.168.1.2

3. If the connection is successful, replies such as 'Reply from 192.168.1.2' will be displayed.
4. Repeat the ping test between the other computers to verify LAN connectivity.

Result

A LAN was successfully setup and configured using a network switch and Ethernet cables. The computers were assigned unique IP addresses and communication between the connected computers was verified using the ping command.

Precautions

- Ensure that Ethernet cables are properly connected to both the computers and the switch.
- Assign a unique IP address to every computer in the LAN.
- Ensure that all computers use the same subnet mask.
- Check that the Ethernet adapter is enabled before testing the connection.
- Do not disconnect network cables while the LAN is being tested.